



Exploring activity-induced dental modifications in medieval Pieve di Pava (central Italy, 10th-12th centuries AD)

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ABSTRACT

Objective: This study aims to investigate the following aspects: i) presence of activity-induced dental modifications (AIDMs) in a medieval population from Pieve di Pava (Siena, central Italy); ii) sex-specific and age-specific distribution of AIDMs in the sample; iii) potential correlations between AIDMs and the traditional activities and cultural habits of rural communities in medieval Italy.

Design: The permanent teeth of 204 individuals buried at Pieve di Pava (10th-12th centuries AD) were systematically examined in order to assess the distribution of five types of AIDM, i.e. lingual surface attrition of the maxillary anterior teeth (LSAMAT), excessive occlusal load, grooving, notching, and chipping.

Results: Prevalence rates of LSAMAT, excessive occlusal load and grooving were low, whereas higher frequencies were recorded for the other types of AIDMs. Prevalence was higher in the male subsample for all the modifications recorded. Overall, the frequencies of AIDMs increased with age.

Conclusions: The very low prevalence of LSAMAT was thought to be connected to leatherworking in the sample under study, while excessive occlusal load was interpreted as evidence that hard materials requiring preliminary chewing and softening were not regularly processed with teeth. The unique case of interproximal grooving in an old male individual may reflect the use of a toothpick for therapeutic purpose. Conversely, the very high prevalence of notching and chipping was taken as indicative of the processing of wool fibers. The sex-specific distribution of these AIDMs suggests that males performed activities involving the use of teeth more frequently than females.

1. Introduction

Over the last two decades, research on activity-induced dental modifications (AIDMs) has received increasing attention because of its potential for the investigation of aspects pertaining to the diet, oral hygiene, food processing techniques, cultural modifications, working activities, and habits of ancient communities (Berbesque et al., 2012; Irvine, Thomas, & Dietrich-Schoop, 2014; Lorkiewicz, 2011; Molnar, 2008).

Chewing foods is acknowledged to be the principal activity inducing

dental surface modifications in ancient as well as contemporary populations; however, it is not the only process causing an irreversible loss of dental hard tissue (Imfeld, 1996). Indeed, it is generally accepted that dental wear results from the interaction between three main mechanical stresses: attrition, abrasion, and erosion. These three terms are all used to indicate a loss of enamel/dentine caused by the interplay of different factors (Addy & Shellis, 2006; Berbesque et al., 2012; Erdal, 2008; Kieser et al., 2011; Robb, Cruwys, & Smith, 1991; Wallace, 1974).

Attrition is defined as tooth-on-tooth contact during mastication, affecting the occlusal surface of teeth (Imfeld, 1996); abrasion relates to

Abbreviations: AIDMs, Activity-induced dental modifications; LSAMAT, Lingual surface attrition of the maxillary anterior teeth; MX, Maxilla; MD, Mandible.

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friction between the teeth and exogenous, abrasive agents (Addy & Shellis, 2006); while chemical etching and the consequent dissolution of the mineral content of teeth induced by “acid or chelation without bacterial involvement” (e.g., acidic dietary components) is referred to as erosion (Imfeld, 1996). Abrasion patterns encompass an array of dental modifications resulting from task-oriented activities, which may be further categorized as: i) intentional modifications for aesthetic purposes (e.g., abrasion and bleaching) (Alt & Pichler, 1998); ii) intentional modifications to express social identity and cultural affiliations (e.g., ablation, mutilation, and filing) (Durband, Littleton, & Walshe, 2014; Stojanowski, Carver, & Miller, 2014; Willman, Shackelford, & Demeter, 2016); iii) unintentional modifications caused by personal habits (e.g., pipe notch) (Kvaal & Derry, 1996); iv) unintentional modifications related to medical and oral hygiene practices (e.g., use of toothbrush and dental probes) (Frayer & Russell, 1987); and v) unintentional modifications connected with tooth-tool use (i.e. use of teeth as a “third hand”) in occupational activities (e.g., processing of plant fibers for weaving mats and clothing, basket-making, and leatherworking) (Irvine et al., 2014; Lorkiewicz, 2011; Molnar, 2011; Tanga, Quintili, Tiranelli, D'anastasio, & Viciano, 2016). Behaviorally induced dental defects have been successfully used as a proxy for reconstructing habitual actions, social division of labor, and types and modes of production (Frayer & Russell, 1987; Irvine et al., 2014; Lorkiewicz, 2011; Molnar, 2011; Tanga et al., 2016). According to Scott (1997), ancient populations, as well as modern populations that cannot rely on diversified and complex toolkits, were accustomed to using hands and teeth to perform a variety of tasks. Teeth, in particular, could be utilized as pliers and tools for stripping and tearing, thanks to their strength and shape (Scott, 1997). Nevertheless, scholars advocate caution when relating dental modifications to specific activities (Erdal, 2008; Lorkiewicz, 2011; Molnar, 2008). In the case of chipping, for instance, traumatic and/or dietary etiologies have been suggested in addition to occupational causes (e.g., housework and crafts) (Bonfiglioli, Mariotti, Facchini, Belcastro, & Condemi, 2004). Accordingly, the combined use of ethnographic, archaeological, and ecological data provides the most conclusive approach for a critical examination of unusual dental wear patterns to infer occupational activities. Ethnographic sources have often played a crucial role in the identification and interpretation of activity-induced dental wear (e.g., Brown & Molnar, 1990; Erdal, 2008). Indeed, traditional modes of production, generally abandoned as a result of technological innovation, survive in specific rural contexts or are at least remembered by some living populations (Erdal, 2008). This wealth of information can help to elucidate the ways in which past groups used

their teeth for work-related purposes. Additionally, archaeological and ecological data provide useful insights into sociocultural contexts and the exploitation of aquatic or terrestrial resources that may have been available in antiquity, facilitating the identification of causative agents responsible for the occurrence of relevant dental defects (e.g., plant fibers, animal skin) (e.g., Irish & Turner, 1997; Turner & Machado, 1983). Unfortunately, this specific category of occupational markers has very rarely been explored in Italian osteological collections, particularly as far as the medieval period is concerned.

The aim of the present study is threefold: (i) to document AIDMs in a medieval rural population from Italy, contributing to current knowledge of this period from a bioarchaeological perspective; (ii) to assess the sex-specific and age-specific distribution of AIDMs; (iii) to explore potential correlations between extramasticatory wear patterns and the traditional activities and cultural habits of Italian rural communities during the medieval period.

2. Materials and methods

2.1. Archaeological context

Pieve di Pava is a rural site located in southern Tuscany (central Italy, Fig. 1), in the hilly landscape of Val d'Orcia and the Crete Senesi, near the city of Siena (Campana, Felici, & Marasco, 2008). The archaeological excavation of the site was conducted between 2004 and 2014 and documented a complex stratigraphic sequence (Felici, 2016a).

The first known traces of occupation at Pieve di Pava date back to Roman times, and more precisely to the 2nd-3rd centuries AD, when a rural *villa* was built in the area. This building was still in use during Late Antiquity (5th century AD), when its interior space was repurposed to host burials, as testified by a capuchin tomb marking its functional shift from residential to funerary (Felici, 2016a, 2016b). Between the end of the 5th century and the beginning of the 6th century AD, a large paleo-Christian church or *plebs* (32 × 20 m) dedicated to S. Pietro was erected in the space previously occupied by the *villa*. The first mention of this religious building in historical documents dates back to 715 CE. During the Early Middle Ages (7th-8th centuries AD), the church underwent a series of structural interventions, which mostly reduced its size; simultaneously, a number of inhumations in earth-dug graves with lithic elements were placed in the area near its entrance (Felici, 2016a, 2016b).

The privileged position of the burials surrounding the entrance to the *plebs* of S. Pietro suggests the existence of a social elite that might have

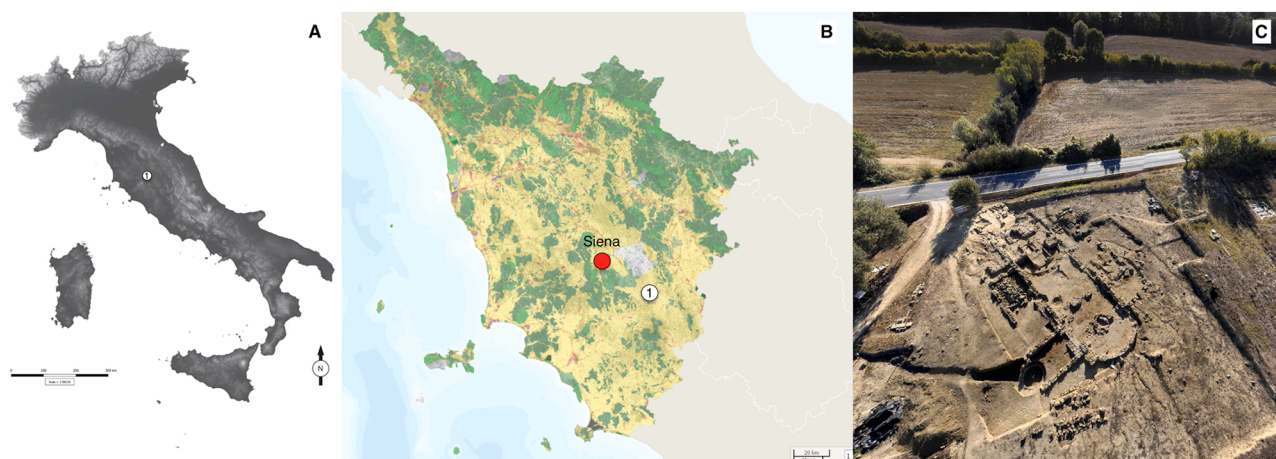


Fig. 1. (a) Map of Italy showing the location of the archaeological site of Pieve di Pava; (b) map of Tuscany showing the location of Pieve di Pava and the main city of Siena; (c) aerial photo of Pieve di Pava (from Felici, 2016b).

Maps adapted from public domain wms data available at <http://www.pcn.minambiente.it/mattm/servizio-wms/> and <https://www.regione.toscana.it/-/geoscopio-wms> under a CC BY license, with permission from Geoportale Nazionale and GEOscopio Regione Toscana, respectively

commissioned the renovation of the church in the 7th century AD (Campana, Felici, Fornaciari, Mongelli, & Vitiello, 2015). These burials can reasonably be ascribed to members of Longobard aristocracies, who were present in the territory of Siena as exponents of both military hierarchies and landed nobility with a strong interest in the Christianization process (Tabacco, 1973). The attribution of the sumptuous jewelry adorning a female skeleton to the Longobard period has been confirmed by radiocarbon dating (641–667 CE) (Campana et al., 2015; Felici, 2016b). Moreover, a unique case of secondary burial of a young male individual (Mongelli, Vitiello, Campana, Lubritto, & Fornaciari, 2011) was detected in front of the altar; radiocarbon dating of the osteological material provided a chronological range between 650 and 688 CE (Felici, 2016b). It has been hypothesized that the presence of this privileged burial in such an important position was somehow connected with the increasingly extensive use of the external area of the church for funerary purposes between the 10th and 12th centuries AD (Campana et al., 2015). Indeed, while the *plebs* was significantly reduced in size over this period, the disposition of relevant burials was notably chaotic, with the overcrowding and frequent overlapping of inhumations. Approximately 900 burials, mostly simple earth-dug graves without funerary equipment, were brought to light during the excavations carried out at the site (Felici, 2016b). However, only a relatively small number of individuals were submitted for anthropological and paleopathological analyses. As for the identification of the community buried in Pieve di Pava, although archaeological investigations have not managed to locate the settlement associated with the church yet (Campana et al., 2009; Felici, 2012), it can be suggested that the 10th- to 12th-century graves dug in its proximity belonged to common people and local devotees under the ecclesiastical influence and control of the *plebs* of San Pietro in Pava (Felici, 2016a).

The religious role of the *plebs* waned in the 11th century AD, and the building was then abandoned in favor of the church of Santa Maria, erected on a nearby hill, in the 12th century AD. Despite its abandonment, the church of S. Pietro retained its funerary function at least until the mid-13th century AD (Felici, 2016b), as documented by the radiocarbon dating of osteological samples (1160–1222 CE) (Campana et al., 2007).

2.2. Sample selection

A sample of 204 individuals fulfilling specific criteria was selected for the purpose of this study. In particular, sample selection was based on the inclusion of: i) individuals dating to the 10th–12th centuries AD; ii) individuals whose biological sex could be confidently estimated as male or female; iii) adults (> 20 years old) (Falys & Lewis, 2011). The analysis involved the assessment of all the teeth available per individual.

2.3. Biological sex and age-at-death estimation

Biological sex was estimated on the basis of sexually dimorphic cranial and pelvic traits (Acsádi & Nemeskéri, 1970; Ferembach, Schwidetzky, & Stloukal, 1980). Age-at-death estimation relied on the assessment of morphological changes to the pubic symphysis (Brooks & Suchey, 1990), the auricular surface of the ilium (Buckberry & Chamberlain, 2002; Lovejoy, Meindle, Pryzbeck, & Mensforth, 1985), and the sternal rib ends (Burns, 2013). In order to enable statistical comparison, each individual was assigned to a specific age group (young adult: 20–24 years, middle adult: 25–34 years, mature adult: 35–44 years, old adult: 45+ years), following Işcan and Steyn (2013).

2.4. Assessment of activity-induced dental modifications

The evaluation of AIDMs was carried out both macroscopically and using a microscope (Photomicroscope Leica WILD M400, Heerbrugg, Switzerland; 6.3–32.0 magnification). The morphology and severity of each category of dental defects were assessed separately.

Lingual surface attrition of the maxillary anterior teeth (LSAMAT), i.e. advanced lingual wear involving the upper anterior teeth without lingual or labial surface wear on any of the corresponding mandibular teeth, was exclusively recorded as present/absent (Turner & Machado, 1983). Excessive occlusal load, which can be described as extreme, non-homogenous, and frequently oblique wear leaving little to no enamel on tooth surfaces in the molar and premolar regions, was exclusively recorded as present/absent (Molnar, 2008). Grooving, i.e. the presence of tubular channels running in a linguolabial or mesiodistal direction across the wear plane of the occlusal surface of teeth (occlusal grooves) or in a linguolabial direction on the interproximal surfaces of teeth, at the cemento-enamel junction (interproximal grooves), was exclusively recorded as present/absent. Moreover, interproximal grooves were evaluated according to the criteria described by Frayer and Russell (1987). The severity of grooving was assessed using a precision 0.1 mm electronic caliper under a photomicroscope. Notching, i.e. the presence of ante-mortem indentations exhibiting greater width than length on the occlusal edge of teeth, was scored according to the three-grade scale proposed by Bonfiglioli et al. (2004). Following their classification, grade 1 was assigned to slight, superficial indentations affecting only the enamel; grade 2 was assigned to wider and deeper indentations with polished dentine; and grade 3 was assigned to very deep and equally wide depressions with heavily polished dentine. Lastly, chipping, i.e. the ante-mortem crack or break of a portion of the tooth crown involving enamel or enamel and dentine and mostly occurring at the occlusal edges of teeth, was scored following the three-grade scale by Bonfiglioli et al. (2004). Accordingly, grade 1 was assigned to slight cracks or fractures (0.5 mm), or larger but superficial enamel flake loss; grade 2 was assigned to square irregular lesions (1 mm) with the enamel more deeply involved; and grade 3 was assigned to cracks bigger than 1 mm involving enamel and dentine, or large, very irregular fractures that could destroy the tooth.

2.5. Statistical analysis

Absolute frequencies of AIDMs were estimated as crude prevalence rates (number of affected/observed individuals) and true prevalence rates (number of affected/observed teeth), and were compared in order to highlight differences in their distribution between the sexes.

A Chi-square test was run to compare sex-specific frequencies and verify whether potential differential patterns of distribution were statistically significant. Specifically, statistical comparisons between the sexes focused exclusively on the maxillary anterior teeth in the case of LSAMAT, while the sex-specific distributions of notching and chipping in the maxillary anterior teeth, the maxillary posterior teeth, the mandibular anterior teeth, and the mandibular posterior teeth were compared separately. Loglinear analysis was used to explore the relationship between the distribution of the five AIDMs and age-at-death.

Furthermore, true prevalence rates were compared in order to explore variations in the severity distribution of chipping and notching across the age groups. Since the study involved work on ordinal and non-normally distributed variables, the normality of sample distribution was tested through the Kolmogorov-Smirnov test. The Kruskal-Wallis test and the resulting *H*-statistic were used to compare medians between the different age groups considered. Statistical analysis was performed using IBM® SPSS® Statistics 25.0, with significance set at $p < 0.05$.

3. Results

The sample under study consisted of 90 males and 114 females, for a total of 204 individuals. The majority of the individuals fell in the mature adult age range; females could mainly be placed in the middle adult and mature adult groups, while males mostly belonged to the mature adult and old adult groups. Young adults of both sexes were underrepresented (Table 1).

Crude prevalence rates and true prevalence rates of extramasticatory

Table 1
Sex and age composition of the sample.

	Young Adulthood (20–24)	Middle Adulthood (25–34)	Mature Adulthood (35–44)	Old Adulthood (45 +)	Total
Males	7	16	46	45	90
Females	16	32	36	6	114
Total	23	48	82	51	204

dental wear were calculated for both sexes and compared in order to garner insights into the sex-specific distribution of AIDMs. A total of 4222 teeth were available for observation (1856 maxillary teeth and 2366 mandibular teeth).

LSAMAT (Fig. 2) was detected in 16.2% of the individuals observed (33/204), and males were found to be overall more affected than females (23/90, 25.5% vs 10/114, 8.7%) (Table 2). This difference was statistically significant ($\chi^2 = 11.595$; $p < 0.01$). As for the true prevalence of LSAMAT (Table 3), males showed higher rates for all the maxillary anterior teeth compared to females. The comparison of the frequencies of affected maxillary incisors and canines between the sexes revealed significant differences in relation to the first incisor ($\chi^2 = 19.951$, $p < 0.01$), the second incisor ($\chi^2 = 16.823$; $p < 0.01$), and the canine ($\chi^2 = 8.864$; $p < 0.01$).

Excessive occlusal load (Fig. 3) was recorded only in two individuals (2/204, 0.9%), one male and one female (Table 2). Five teeth exhibited this unusual dental wear pattern: they were all first and second molars, two from the maxilla and three from the mandible (Table 3). In this case, no statistical test was performed due to the small amount of representative data.

Grooving (Fig. 4) was observed in a single individual, an old adult male (Table 2). One of the two grooves detected was located between the left maxillary central and lateral incisors; the second groove was visible on the mesial edge of the right maxillary lateral incisor and had probably also involved the central incisor, mirroring the defect observed on the left side of the maxilla. Unfortunately, however, this tooth was not observable owing to post-mortem loss. Both grooves exhibited a smooth, polished surface and were roughly semicircular, with a diameter of approximately 2–4 mm.

Notching (Fig. 5) was recorded in 46.1% of the individuals observed (94/204), and males were more affected than females (54/90, 60.0% vs 40/114, 35.1%) (Table 2). This difference was statistically significant ($\chi^2 = 12.562$, $p < 0.01$). As in the case of LSAMAT, males exhibited a higher true prevalence rate of notching. The comparison of the frequencies of affected teeth between the sexes revealed significant differences in relation to both dental arches (maxillary anterior teeth, $\chi^2 = 40.979$, $p < 0.01$; maxillary posterior teeth, $\chi^2 = 11.732$, $p < 0.01$; mandibular anterior teeth, $\chi^2 = 39.592$, $p < 0.01$; mandibular posterior

teeth, $\chi^2 = 4.554$, $p = 0.032$).

Chipping (Fig. 6) was noted in 53.4% of the individuals observed (109/204); as for the other types of AIDMs, males were found to be more affected than females (63/90, 70.0% vs 46/114, 40.3%). Also in this case, the difference was statistically significant ($\chi^2 = 17.768$, $p < 0.01$). Upper and lower teeth exhibited the highest rates of chipping in the male subsample. Significant differences between the sexes were detected in the maxillary anterior teeth ($\chi^2 = 6.265$, $p = 0.012$) and the maxillary posterior teeth ($\chi^2 = 10.160$, $p < 0.01$), while no differences emerged in relation to the mandibular anterior teeth ($\chi^2 = 1.609$, $p = 0.204$) and the mandibular posterior teeth ($\chi^2 = 2.973$, $p = 0.084$).

Concerning the distribution of the five types of AIDM by age (Table 4), LSAMAT appeared with similar frequencies in the middle adult, mature adult, and old adult age groups ($p = 0.158$), while no evidence of this AIDM was noted in the dentitions of young adult individuals. Similarly, notching was documented in all the age groups, with no statistically significant differences ($p = 0.540$). Chipping was found mainly in young adults (14/23, 60.9%), with frequencies decreasing slightly in old adults (25/51, 49.0%); however, also in this case, the difference was not statistically significant ($p = 0.820$). Since excessive occlusal load and grooving were only observed in two and one individuals, respectively, no specific age patterns could be reconstructed for these two types of AIDM.

With regard to the severity distribution of notching by age (Fig. 7), mild forms (grade 1) prevailed in young adults (24/25, 96.0%), gradually decreasing toward mature adulthood (63/107, 68.3%). Conversely, the prevalence rates of grade 2 and grade 3 notching increased from the young adult group (1/25, 4.0% grade 2) to the old adult group (13/60, 17.5% grade 2 and 7/60, 12.3% grade 3). The difference in the rank distribution of this AIDM between age groups was statistically significant (H -statistic 9.929, $p = 0.019$). Concerning the severity distribution of chipping by age (Fig. 8), the prevalence rate of its most severe forms (grade 3) gradually increased with age (young adults 2/48, 4.2%; middle adults 4/71, 5.6%; mature adults 15/189, 7.9% and old adults 9/83, 10.8%), while moderate forms (grade 2) were almost equally distributed between the four age groups considered (20.8–19.3%). Lastly, mild forms of chipping (grade 1) decreased slightly from the young adult group (36/48, 75.0%) to the mature adult group (133/189, 70.4%). No statistically significant differences were detected in the rank distribution of this AIDM between age groups (H -statistic 0.754, $p = 0.861$).

4. Discussion

Based on the evaluation of AIDMs in a sample of 204 individuals from the archaeological site of Pieve di Pava, it was possible to observe that: i) males presented higher frequencies of three dental defects (LSAMAT, notching, chipping) compared to their female counterparts; ii) the most recurrent types of AIDM were notching (46.1% of the individuals) and chipping (53.4% of the individuals); iii) only one mature male showed evidence of interproximal grooving (0.4%); iv) only two individuals, a female and a male, displayed excessive occlusal load (0.9%); v) younger individuals exhibited very low prevalence rates of AIDMs compared to the older age groups.

LSAMAT has been identified in a number of osteological collections around the world (Berbesque et al., 2012; Harper & Fox, 2008; Irish & Turner, 1987; Liu, Zhang, Wu, & Zhu, 2010; Pechankina, Benfer, & ZhiJun, 2002; Rodrigues, Silva, Matias, & Santos, 2021; Saul & Saul, 2006; Tanga et al., 2016; Turner & Machado, 1983; Watson & Haas, 2017). There is almost unanimous consensus on the etiology of this unusual wear pattern, which has been widely related to the peeling and shredding of plant fibers. It must be noted, however, that the softening of animal skin as part of the leatherworking process has been alternatively postulated as a potential cause for this AIDM. Lous (1970), for instance, included the photo of an Inuit woman using her incisors to soften a piece of animal skin in his paper. Both operations, connected to the



Fig. 2. Evidence of LSAMAT on the maxillary anterior teeth of a middle adult male buried in US 13046.

Table 2

Sex-specific crude prevalence rates of LSAMAT, excessive occlusal load, grooving, notching, and chipping. n: number of individuals affected; N: total number of individuals observed for each sex. Statistically significant differences between males and females are indicated with an asterisk (*); significance set at $p < 0.05$.

		LSAMAT*	Excessive occlusal load	Grooving	Notching*	Chipping*
Males	n/N	23/90	1/90	1/90	54/90	63/90
	%	25.5	1.1	1.1	60.0	70.0
Females	n/N	10/114	1/114	0/114	40/114	46/114
	%	8.7	0.8	0	35.1	40.3%
Total	n/N	33/204	2/204	1/204	94/204	109/204
	%	16.2	0.9	0.4	46.1	53.4

preparation of plant fibers and animal sinew, respectively, represent the preliminary stages in the processing of raw materials for the production of cordage, nets, baskets, or clothing in ancient and living populations (Pechankina et al., 2002). Concerning the abrasive agent, namely the species of plants that may be associated with evidence of LSAMAT, Turner and Machado (1983) suggested its possible identification with manioc, a starchy plant rich in carbohydrates, which was widely cultivated by the people of Corondò (Brazil) in the Archaic period. To corroborate this hypothesis, they emphasized that these dental defects occurred on the same teeth affected by caries (Turner & Machado, 1983). Watson and Haas (2017), instead, did not recognize any correlation between the rate of caries and that of LSAMAT in a coeval osteological collection from the Lake Titicaca Basin (Peru). Thus, they proposed that “one or more progenitor species of endemic tubers”, such as potatoes, which had been domesticated from wild species in the south-central Andes and are less sugary than tropical tubers, might have been responsible for the abrasion patterns observed on teeth (Watson & Haas, 2017). Sugar cane was associated with the high rates of LSAMAT recorded by Irish and Turner (1997) in an early 20th-century skeletal series from Senegal. Indeed, the diameter of this plant, ranging between 2.5 and 5.0 cm, was deemed compatible with the dimensions of the abrasions observed (Irish & Turner, 1997). As it emerges from these three examples, knowledge of the geographical setting combined with ethnographical records is crucial for a well-supported identification of abrasive agents connected with LSAMAT. The distribution of this type of AIDM in the sample from Pieve di Pava would suggest that a limited number of individuals (16.1%), mostly males, regularly employed their teeth to remove hard components from plants. Unfortunately, these plants cannot be confidently identified owing to the lack of archaeobotanical evidence from the site. Additionally, it has to be taken into consideration that the rural vocation of this community also found expression in economic activities other than crop production, and particularly livestock farming, as testified by archaeologically documented remains of ovicaprids and pigs at Pieve di Pava (Campana et al., 2008; Ricci et al., 2012; Riccomi, 2021). Accordingly, this AIDM might also have originated from the performance of repeated sets of movements aimed at softening animal skin for the processing of leather. The identification of activities in which the members of the community of Pieve di Pava might have been engaged on a regular basis paves the way for additional considerations on labor division at the site. Indeed, archival sources bear witness to the perceived centrality of leatherworking in the economy of the province of Siena during the medieval period (Balestracci, 1999). The first statutory codification of leatherworking as a profession is dated to 1329 (i.e. *Formula di giuramento dei sottoposti all'arte dei calzolari*, [oath for cobblers]) and this document is particularly important as it records a remarkable number of shoemakers, tanners, and cloth sellers in activity between 1240 and 1270 CE (Balestracci, 1999; Banchi, 1877). Moreover, it states that leatherworkers undertook their apprenticeship at only 14 years of age (Balestracci, 1999) and regulates access to corporations, ensuring, for instance, that sons could take over their fathers' activity at their death (Banchi, 1877). Based on the information provided by this official document, it can be inferred that the leatherworking profession was reserved for the male component of society, as documented elsewhere in Italy (Rodighiero & Carrara, 2013). Regarding age patterns, the

preponderance of affected individuals in the mature adult and old adult age groups would seem to indicate that many years of repetitive mechanical stresses were necessary to produce this peculiar type of AIDM.

Excessive occlusal load was observed only in two individuals. Owing to the scarcity of available data, any interpretation would be hazardous. However, it must be noted that excessive occlusal load may be due to both masticatory mechanisms and extramasticatory behaviors. Indeed, molars, which are the most frequently affected teeth, can serve to soften plants or sinew as a preliminary step in the processing of plant and animal fibers or simply as part of the mastication process (Kieser et al., 2011; Mair, Stolarski, Vowles, & Lloyd, 1996; Molnar, 2008).

The two cases of interproximal grooving identified at Pieve di Pava were observed in a single individual. Interestingly, the grooves were not located at the cemento-enamel junction, which is the most recurrent position reported in the literature (Wallace, 1974), but immediately above the dental neck, on the mesiodistal surfaces of the first and second incisors both on the right and left sides of the maxilla (Fig. 3). Grooving has been detected in a number of osteological assemblages mainly dating back to prehistoric times (Estalrich, Alarcón, & Rosas, 2017; Formicola, 1988; Frayer & Russel, 1987; Frayer, 1991; Irvine et al., 2014; Lozano, Subirà, Aparicio, Lorenzo, & Gómez-Merino, 2013; Minozzi, Manzi, Ricci, Di Lernia, & Borgognini Tarli, 2003; Speduti et al., 2018; Waters-Rist, Bazaliiskii, Weber, Goriunova, & Katzenberg, 2010), while the medieval (Erdal, 2008; Rodrigues et al., 2021) and modern historical periods have rarely been investigated (Brown & Molnar, 1990; Ubelaker, Phenice, & Bass, 1969). The insertion of inflexible instruments between the teeth for palliative and therapeutic purposes or in relation to habitual idiopathic tooth-picking is often cited among the most plausible causes for this type of AIDM (Berryman, Owsley, & Henderson, 1979; Ubelaker et al., 1969; Ungar, Grine, Teaford, & Perez-Perez, 2001). Furthermore, additional causative factors, such as the processing of sinew and plant fibers (e.g., hemp, *Cannabis* sp.) should be taken into consideration (Brown & Molnar, 1990), especially in the case of occlusal grooves in the anterior dentition (Larsen, 1985; Schulz, 1977; Waters-Rist et al., 2010). The majority of cases of grooving presented in the literature concern the posterior teeth (e.g., Brown & Molnar, 1990; Formicola, 1988; Frayer & Russel, 1987), while only few studies record a different location of the defects within the anterior teeth (e.g., Berryman et al., 1979). The former include a case reported by Trombley and colleagues (2018), who detected a semi-circular interproximal groove on the left maxillary second premolar of an individual from the late medieval site of Villamagna, Italy (1350–1500 CE). Based on the observation of parallel micro-striae oriented in a bucco-lingual direction and on the tubular shape of the defect, the groove was ascribed to “an exogenous implement abrading with the root surface for hygienic or palliative reasons” (Trombley, Agarwal, Beauchesne, & Goodson, 2018). The presence of grooves on both the anterior and posterior teeth is discussed in a work by Berryman and colleagues (1979), who noted these dental defects between the right first and second maxillary incisors, and between the right first, second, and third maxillary and mandibular molars of an individual from the site of Larson (Dakota) (1750–1785 CE). In this case, the use of dental probes was identified as the only possible cause behind the localized abrasions. The location of these defects on the anterior dentition is consistent with that recorded at Pieve di Pava. While there was no evidence of dental

Table 3
True prevalence rates of LSAMAT, excessive occlusal load, grooving, notching, and chipping by sex. n: number of teeth affected; N: total number of teeth observed; NA: not available. Statistically significant differences in the affected maxillary incisors and canines between the sexes are indicated with an asterisk (*); statistically significant differences in the affected maxillary anterior teeth, maxillary posterior teeth, mandibular anterior teeth, and mandibular posterior teeth (considered separately) between the sexes are indicated with two asterisks (**); significance set at $p < 0.05$.

	LSAMAT				Excessive occlusal load				Grooving				Notching**				Chipping**				Total
	Male		Female		Male		Female		Male		Female		Male		Female		Male		Female		
	n/N	%	n/N	%	n/N	%	n/N	%	n/N	%	n/N	%	n/N	%	n/N	%	n/N	%	n/N	%	
MX	36/113	31.9	9/112	8.0 *	0/113	0	0/112	0	2/113	1.7	0/112	0	22/113	19.5	5/112	4.5	52/113	46.9	46/112	41.1	225
I ¹	31/124	25.0	7/119	5.9 *	0/124	0	0/119	0	2/124	1.6	0/119	0	21/124	16.9	2/119	1.7	47/124	37.9	25/119	21.0	243
I ²	11/137	8.0	1/139	0.7 *	0/137	0	0/139	0	0/137	0	0/139	0	15/137	10.9	1/139	0.7	35/137	25.5	30/139	21.6	276
C	3/139	2.2	0/128	0	0/139	0	0/128	0	0/139	0	0/128	0	4/139	2.9	0/139	0	15/139	10.8	9/128	7.0	267
P ¹	2/127	1.6	0/131	0	0/127	0	0/131	0	0/127	0	0/131	0	7/127	5.5	0/131	0	9/127	7.1	4/131	3.1	258
P ²	1/95	1.1	0/122	0	0/95	0	1/122	0.8	0/95	0	0/122	0	1/95	1.1	1/122	0.8	7/95	7.4	3/122	2.5	217
M ¹	0/91	0	0/118	0	0/91	0	1/118	0.8	0/91	0	0/118	0	0/91	0	0/118	0	3/91	3.3	1/118	0.8	209
M ²																					
M ³	0/75	0	0/86	0	0/75	0	0/86	0	0/75	0	0/86	0	1/75	1.3	0/86	0	3/75	4.0	0/86	0	161
Total																					1856
MD																					
I ₁	NA		NA		0/150	0	0/131	0	0/150	0	0/131	0	26/150	17.3	8/131	6.1	15/150	10	12/131	9.2	281
I ₂	NA		NA		0/175	0	0/140	0	0/175	0	0/140	0	31/175	17.7	0/140	0	17/175	9.7	13/140	9.3	315
C	NA		NA		0/184	0	0/154	0	0/184	0	0/154	0	12/184	6.5	1/154	1.6	17/184	9.2	6/154	3.9	338
P ₁	NA		NA		0/194	0	0/157	0	0/194	0	0/157	0	7/194	3.6	2/157	1.3	7/194	3.6	5/157	3.2	351
P ₂	NA		NA		0/178	0.5	0/145	0	0/178	0	0/145	0	4/178	2.2	1/145	0.7	0/178	0	1/145	0.7	323
M ₁	NA		NA		1/143	0.6	1/127	0.7	0/143	0	0/127	0	6/143	4.2	2/127	1.6	5/143	3.5	0/127	0	270
M ₂	NA		NA		0/142	0	1/133	0.7	0/142	0	0/133	0	6/142	4.2	1/133	0.8	4/142	2.8	0/133	0	275
M ₃	NA		NA		0/108	0	0/105	0	0/108	0	0/105	0	1/108	0.9	0/105	0	1/108	0.9	1/105	0	213
Total																					2366

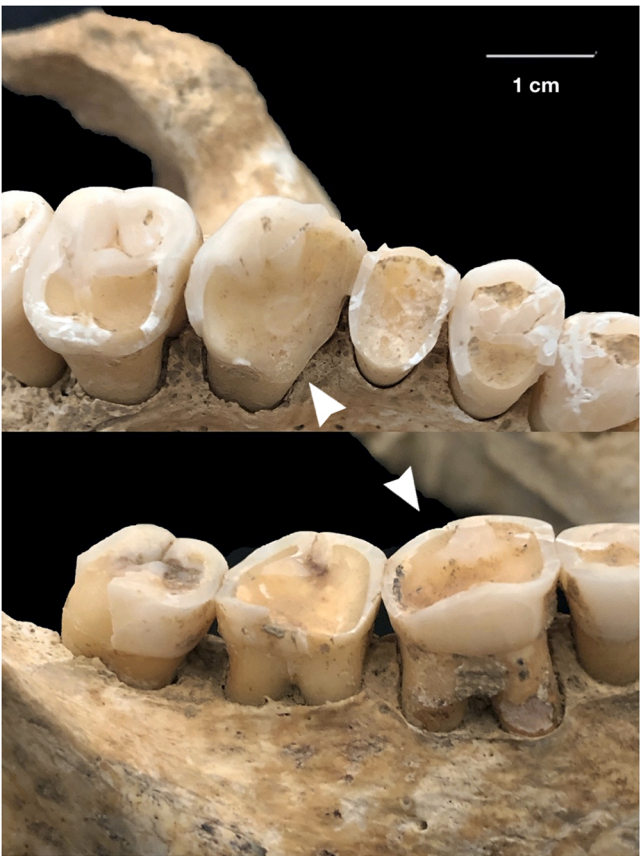


Fig. 3. Evidence of excessive occlusal load on the maxillary (top) and mandibular (bottom) molars of a middle adult female buried in US 4791.

caries and abscesses in the individual from this site, periodontal disease in the form of slight alveolar resorption (around 2 mm) with a ragged and porous appearance of the alveolar margin (Ogden, 2008) in association with the affected incisors (Fig. 4) was observed. Therefore, it is suggested that the most plausible cause for the presence of interproximal grooving in this individual could be identified with the use of repetitive toothpicks as a palliative treatment against periodontal disease. Nevertheless, it cannot be ruled out that the use of toothpicks was additionally prompted by the need to remove food residues from the teeth; indeed, the accumulation of dietary grit may lead to periodontal disease (Berryman et al., 1979).

Concerning notching, the pattern of distribution of this AIMD in the sample from Pieve di Pava is in line with data reported in the literature for contemporary and archaeological contexts: the occurrence of notching was limited to the anterior teeth and its sex-specific distribution showed a predominance in male individuals. The paucity of data on extramasticatory dental wear in the medieval period affects the comparability of these results in terms of prevalence rates. Cruwys and colleagues (1992) found dental notching in eight Anglo-Saxon male skulls belonging to a larger British collection (1029 skulls) dated between the Neolithic and the late medieval period. The practice of sliding a thin thread of sinew between the incisors was suggested to be the cause of this AIMD in the individuals observed (Cruwys, Robb, & Smith, 1992). Scott and Jolie (2008) recorded evidence of notching on the anterior teeth of two medieval Norse groups from Greenland. The crude prevalence rate of this AIMD was estimated to be between 7.1% and 12.1%, and the processing of wool fibers was suggested as its most likely cause (Scott & Jolie, 2008). The prevalence rates recorded in the skeletal remains from Pieve di Pava are higher than in the aforementioned studies. However, Scott and Jolie underline that their results may be an underestimate, particularly as far as slight expressions of this type of AIMD are



Fig. 4. Evidence of interproximal grooving on the maxillary first and second incisors of an old adult male buried in US 8110. The corresponding tooth sockets exhibit alveolar resorption (arrows). A second groove is visible on the mesial surface of the right second incisor (red circle). Magnification power: 63 X.

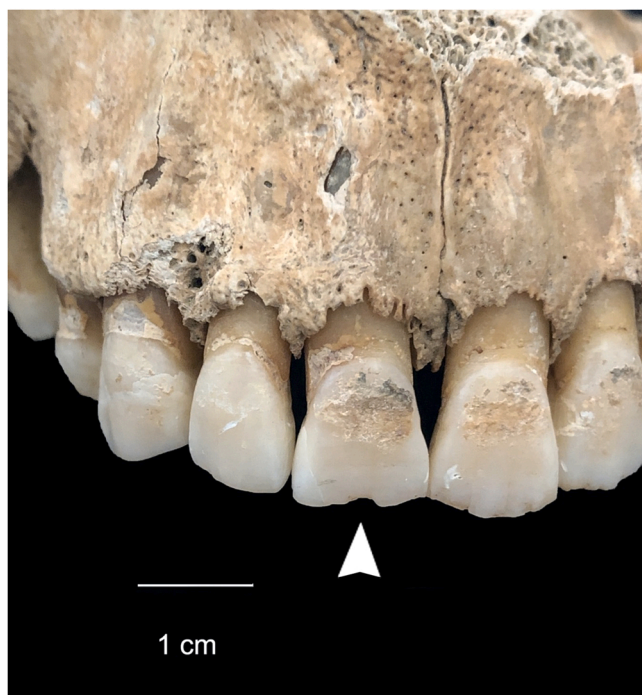


Fig. 5. Evidence of notching on the right maxillary first incisor of a mature adult male buried in US 8670.

concerned, since no scoring system was adopted (Scott & Jolie, 2008). Regardless of the geographical or chronological background of the skeletal collections examined, this dental defect is generally attributed



Fig. 6. Evidence of chipping on the right maxillary first incisor of a mature adult male buried in US 8201. Magnification power: 63 X.

to the habitual holding of small bone objects such as needles or pins between the teeth, or, alternatively, to the stripping of sinew or processing of plant fibers (Bonfiglioli et al., 2004; Lorkiewicz, 2011; Scott & Jolie, 2008; Tanga et al., 2016; Waters-Rist et al., 2010). In a recent study conducted by Rodrigues et al. (2021) on a medieval sample from Santarém (Portugal), the presence of notching was documented in 39.5% of individuals (17/43), with similar frequencies for both sexes. Two different interpretations of this distribution pattern were proposed by the authors: i) the habit of holding wooden sticks between the occlusal surfaces of two corresponding teeth; ii) the consumption of hard-shelled nuts (e.g., walnuts). Quoting Rodrigues et al. (2021): “constant ingestion of dried seeds and hard shells over a chipped crown could increase the chipping area to form a notch”. The second interpretation is also plausible in relation to Pieve di Pava, as archaeobotanical research has revealed the presence of shell fragments belonging to *Corylus avellana* (chestnut) and *Juglans regia* (walnut) in medieval Tuscany (Buonincontri, Pecci, Di Pasquale, Ricci, & Lubritto, 2017). The most affected individuals in the Pieve di Pava sample fell in the middle adult age range, suggesting that relevant dental defects were the result of long-term processes (Waters-Rist et al., 2010). Concerning the severity distribution of notching (Fig. 7), which was carefully evaluated with the

Table 4

Age-specific crude prevalence rates of LSAMAT, excessive occlusal load, grooving, notching, and chipping (sexes pooled). n: number of individuals affected; N: total number of individuals for each age group.

	LSAMAT		Excessive occlusal load		Grooving		Notching		Chipping	
	n/N	%	n/N	%	n/N	%	n/N	%	n/N	%
Young	0/23	0	0/23	0	0/23	0	8/23	34.8	14/23	60.9
Middle	8/48	16.7	1/48	2.0	0/48	0	25/48	47.9	26/48	54.2
Mature	16/82	19.5	1/82	1.2	0/82	0	38/82	46.3	44/82	53.7
Old	9/51	17.6	0/51	0	1/51	2.0	23/51	45.1	25/51	49.0

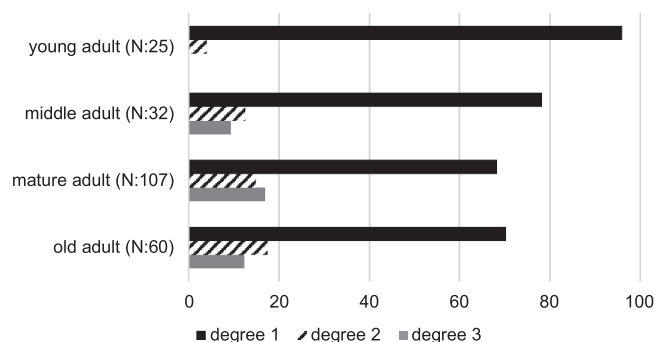


Fig. 7. Severity distribution of notching by age group. N: total number of teeth affected.

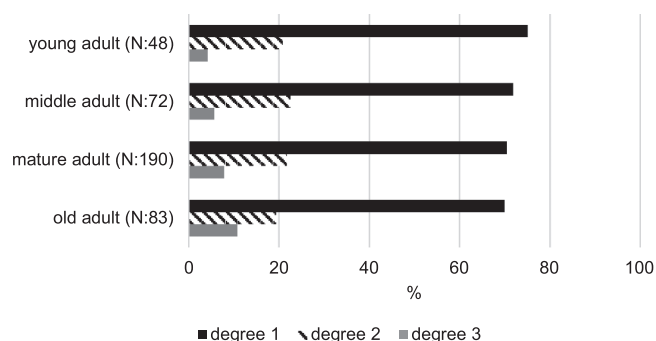


Fig. 8. Severity distribution of chipping by age group. N: total number of teeth affected.

help of a photomicroscope, the increase in rates of moderate (grade 2) and severe (grade 3) defects from the young adult group to the old adult group suggests that both the regular consumption of hard foods from a young age and the repetitive use of teeth as tools might have contributed to the patterns observed (Bonfiglioli et al., 2004, Rodrigues et al., 2021). Furthermore, differences in the sex-specific distribution of notching lead to suppose that repetitive tasks associated with this AIM were predominantly carried out by males. With regard to the types of fiber that could have been processed by the inhabitants of Pieve di Pava, the only information provided by archaeological data is that its rural community was actively involved in ovicaprine husbandry, indirectly suggesting the exploitation of sheep not only for their milk and meat but also for their wool. According to Scott and Jolie (2008), the medieval Norse from Greenland used to treat wool with an alkaline solution before spinning, resulting in an increased abrasiveness of wool fibers. At the end of this treatment, imperfections in the fibers were plucked out with the teeth; this operation was likely carried out while simultaneously using the hands to hold the spindle, so as not to slow down the spinning process (Scott & Jolie, 2008). A similar use of teeth for processing wool can be proposed as a possible cause for the occurrence of notching at Pieve di Pava.

Finally, the high prevalence of chipping in the sample under study was not surprising in consideration of its possible masticatory and

extramasticatory etiologies (Bonfiglioli et al., 2004). Indeed, diet, food preparation, and tooth-tool use are all variables regarded as potentially responsible for the occurrence of crown fractures (Scott & Winn, 2011). Scholarly literature provides evidence of two distinct patterns of distribution of this AIM based on different subsistence activities. Hunter-gatherer populations display high frequencies of chipping in the molar and premolar regions (Bonfiglioli et al., 2004; Irvine et al., 2014; Scott & Winn, 2011), while an “incisor dominant” pattern is recognized in agricultural groups (Scott & Winn, 2011). The greater involvement of the posterior teeth in the first case has been hypothetically related to the chewing of hard, mechanically resistant foods (Belcastro et al., 2007; Fidalgo, Silva, & Porfirio, 2019; Scott & Winn, 2011). Tuner and Cadien (1969), for instance, associated the high rate of chipping (71.9%) in four skeletal series pertaining to prehistoric Inuit groups to the chewing of bones belonging to different animal species, including fish, birds, and land mammals; this argumentation could be corroborated by the numerous cases of avulsion of the anterior teeth detected in the individuals examined. A very similar pattern (97.0% of individuals affected, with a significant preponderance of defects in the molar region) was identified by Scott and Winn (2011) in the prehistoric Inuit of St. Lawrence Island and was related to the consumption of “frozen, tough, grit laden food”. Almost all the individuals (93.9%) from the Epipalaeolithic necropolis of Tavoralt (Morocco) examined by Bonfiglioli et al. (2004) were found to be affected by chipping, with the majority of defects concentrating in the molar region. As in the previous cases, this AIM was tentatively attributed to the chewing of abrasive foods like fruit stones, bones, or parts of snail shells; in this context, the avulsion of the maxillary anterior teeth could have led to a shift of masticatory and non-masticatory functions to the premolar and molar regions (Bonfiglioli et al., 2004). Conversely, the loss of tooth substance in the molar region tends not to be as pronounced in agricultural communities as in non-agricultural groups (Scott & Winn, 2011). This is basically due to differences in subsistence activities and diet, with agricultural communities consuming significantly larger amounts of processed/semi-processed and non-abrasive food. Accordingly, the presence of chipping in the anterior dentitions of individuals belonging to agricultural groups can be more confidently associated with the use of teeth as tools (Scott & Winn, 2011). In the sample from Pieve di Pava, prevalence rates of chipping were high, but not comparable with the frequencies reported in the abovementioned studies. Furthermore, anterior teeth appeared to be more affected than posterior teeth, and the sex-specific distribution of chipping reflected higher frequencies of this type of AIM in males. As in the case of notching, the pattern of distribution of chipping in Pieve di Pava could be related to leatherworking and the processing of vegetable and animal fibers, occupational activities that would have been performed regularly and over very long periods of time. While leatherworking was predominantly a male profession (see above), medieval peasant women were involved in a broad range of daily household tasks that probably required a similar use of teeth, including, for example, sewing, wool and linen production, and cloth-making (Wright, 2020). According to Barber (1994), while weaving appears as a typically female activity in prehistoric contexts, gender roles in textile production became more differentiated during the medieval period (Øye, 2016). Indeed, innovations in textile technology, including the adoption of horizontal looms, and an increasingly

hierarchical organization of production, particularly in urban workshops, prompted a redistribution of tasks at different levels within the textile industry (Øye, 2016). In this context, women would have been involved in textile activities, albeit mostly in a framework of domestic household production and generally without having the juridical identity of professional craftspeople (Øye, 2016).

Accordingly, the lower frequency of chipping observed in females could suggest a quantitative rather than qualitative difference in labor between the sexes. This reconstruction is consistent with other published studies, among which those focusing Iron Age Italy (Tanga et al., 2016); Roman Imperial Italy (Belcastro et al., 2007); and medieval Italy (Belcastro et al., 2007), England (Towle, Davenport, Irish, & De Groote, 2017), Spain (Scott & Winn, 2011), and Portugal (Rodrigues et al., 2021). Conversely, research on Neolithic Sweden (Molnar, 2008) and Bronze Age Portugal (Fidalgo et al., 2019; Silva, Gil, Soares, & da Silva, 2016) documented the opposite trend, with higher rates of chipping in females compared to males.

Based on the preponderance of dental chipping in the young adults of Pieve di Pava (this defect was detected much more frequently in young adults compared to the other types of AIDM), a traumatic/masticatory etiology cannot be ruled out (Bonfiglioli et al., 2004). Indeed, if chipping had ensued from the performance of repetitive tasks using the teeth, higher frequencies of relevant dental defects would have been expected in middle/mature adults (Waters-Rist et al., 2010). The crude prevalence of chipping was similar in middle and mature adults, whereas prevalence rates gradually decreased from the young adult group to the old adult group (Table 4). This trend may be explained in relation to the progressive reduction of the occlusal surface area with age due to physiological dental wear, which might have partially obscured the presence of chipping in older individuals. The evaluation of the severity distribution of chipping across the age groups, which was facilitated by the use of a photomicroscope, suggests that traumatic dental lesions might have occurred as a consequence of the consumption of hard foods even at a young age. Furthermore, the increasing rate of severe forms (grades 2 and 3) of chipping from young adulthood to mature adulthood (Fig. 8) may be ascribed to: i) the effects of the mastication of rough foods on heavily worn teeth, as suggested Bonfiglioli et al. (2004); ii) the cumulative impacts of non-masticatory habits over the life course.

The study of AIDMs has recently gained new impetus, but this line of research has rarely been explored in medieval samples. In this context, the present work provides valuable reference data for comparable studies. Unfortunately, the scarcity of information regarding the environment of Pieve di Pava prevented a confident identification of the agent(s) responsible for the occurrence of the dental defects considered. So far, studies conducted on chronologically or geographically close samples have failed to relate these unusual dental wear patterns to manipulative movements connected with the performance of specific craft activities (Molnar, 2011). The future integration of the results of this study with the microscopic analysis of environmental debris entrapped in dental calculus (e.g., animal and plant fibers) may offer high-definition insights into the materials processed at Pieve di Pava, potentially shedding light on the causal relationship linking AIDMs to traditional activities carried out in medieval rural Tuscany.

5. Conclusions

The AIDMs observed in the medieval skeletal collection from Pieve di Pava bear witness to a greater involvement of males in traditional craft activities, which were partially accomplished through the use of teeth. The presence of LSAMAT in a very small number of individuals, predominantly males, could be interpreted as possible evidence of the processing of leather. The low prevalence of excessive occlusal load recorded on the posterior dentitions of the individuals from Pieve di Pava ruled out the possibility that harder fibers were regularly processed with teeth; otherwise, the molar and premolar regions, which are commonly involved in the chewing and softening of hard fibers, would

have exhibited higher frequencies of unusual dental wear. A unique case of interproximal grooving in an old male individual potentially reflects the use of toothpicks as a therapeutic tool against periodontal disease. Finally, concerning the high prevalence rates of notching and chipping, the processing of rough fibers like wool appears to be one of the most plausible causes for the occurrence of the defects observed, even though a dietary etiology connected with the consumption of nuts and dried seeds cannot be completely ruled out.

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Ethical note

All the human remains included in this study were treated with dignity and respect, in compliance with the highest ethical standards.

CRediT authorship contribution statement

Martina Monaco: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. **Giulia Riccomi:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing – original draft, Visualization. **Simona Minozzi:** Conceptualization, Writing – review & editing. **Stefano Campana:** Acquisition of data, Writing – review & editing. **Valentina Giuffra:** Conceptualization, Writing – review & editing.

Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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